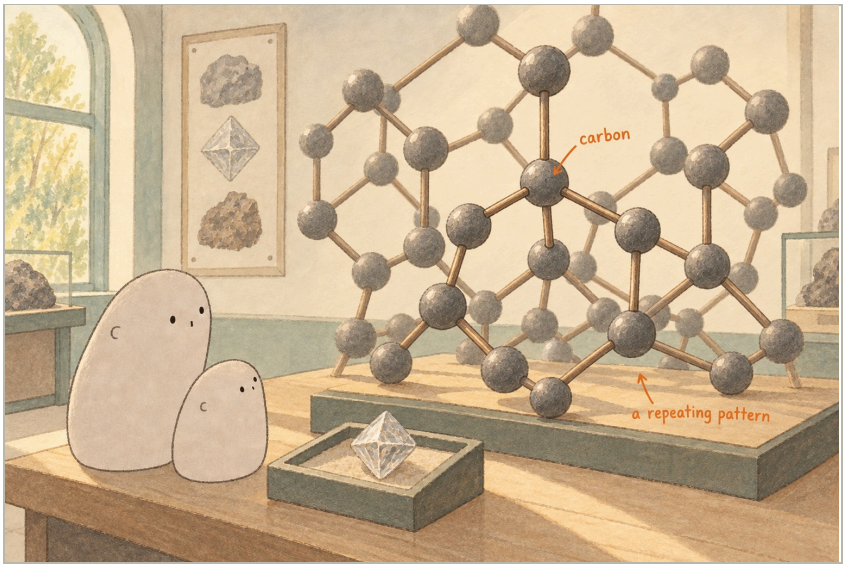
Parts for a pattern

The flash that wouldn't stay. · Curiosity day

Name _____ Date _____



The model reveals a repeating carbon pattern—not a hidden source of light. Real atoms are much too small to see.

PRETEND MODEL SUPPLIES · RODS

Tray	A	B
Rods	18	7

Use these invented supply counts. Do not count the rods in the painting.

1. Tray A has 18 rods. Tray B has 7. How many rods altogether?

18 + 7 = _____

Show your thinking.

2. You have 25 rods. Set 9 aside for later. How many can you use now?

25 − 9 = _____

Show your thinking.

3. Draw 3 separate groups with 4 rods in each. How many rods did you draw?

4 + 4 + 4 = _____

Show your thinking.

These are invented supply counts, not a count of atoms in a diamond. Try drawing a different way to group 12 rods.

WONDER TOGETHER / DAILY

The flash that wouldn't stay.

The flash that wouldn't stay



One leaf. Six steps. A story that keeps going.

GRADE 2

Psst — the leaf is the map.

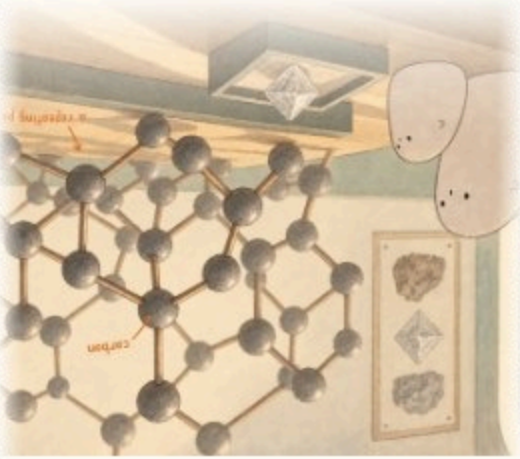
A tiny joke hides in the picture.

A flash winks from a museum case. "The diamond makes light!" says Little Wonderer. Big points to its deep beginning. Heat and huge pressure helped carbon grow into a crystal far underground—without lamp or sun. "Then where does the flash come from?"



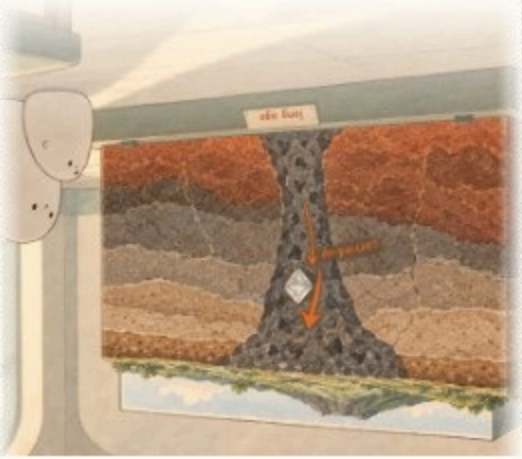
A tiny joke hides in the picture.

"Maybe light hides in the tiny parts," Little says. Balls and rods stand for carbon atoms and their connections. The pattern repeats through a diamond—but it is no lantern. "We found the stone," says Big, "not the flash."



A tiny joke hides in the picture.

Little traces an arrow up. "To meet light, it had to leave the dark!" Long after some diamonds formed, volcanic eruptions carried them toward the surface. "Now it can meet light," says Little. Big nods. "But meeting light is not making it."



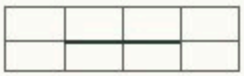
A tiny joke hides in the picture.

A label marked 10 stops Little. Diamond tops the Mohs scratch scale. "So nothing can hurt it?" Big explains that a diamond can still chip or break. "Hard to scratch is not unbreakable," Little says, "and it still doesn't explain the flash."



Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



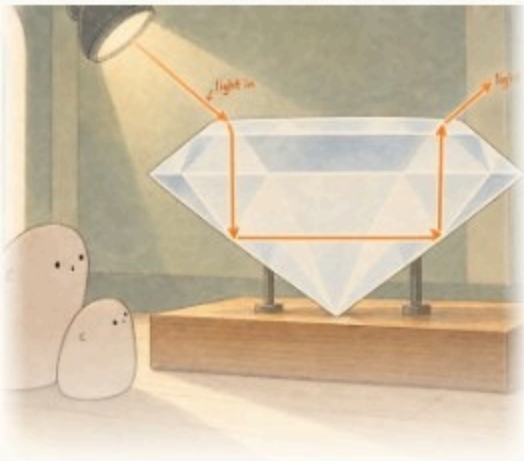
Pictures are simplified and not to scale. Look closely, wonder, and pause whenever you like.

Fold once. Giggle once.



Little shifts. One bright face dims; another flashes. "Found you!" The flash was not stored in the stone. It happens when light, cut facets, and watching eyes meet along the right paths. Little sketches both spots, leaving room for a new question.

A tiny joke hides in the picture.



At the last model, a lamp clicks on. "There!" Little cries. Light enters the diamond, bends, bounces, and comes out again. Cut, flat faces called facets guide its path. Little follows the line. "The diamond sends borrowed light back!"

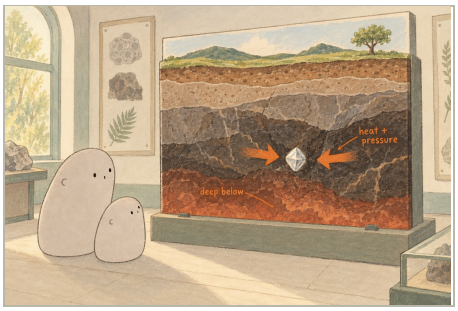
A tiny joke hides in the picture.

Keep a clue from the chase.

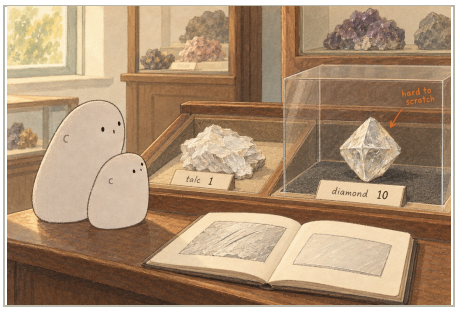
The flash that wouldn't stay. · Curiosity day

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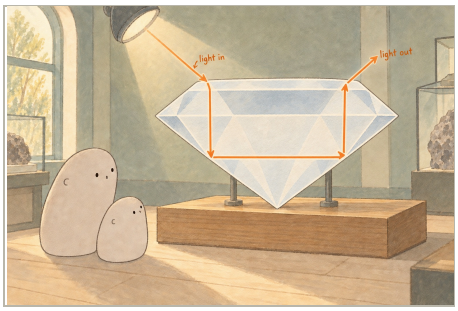
Choose a picture from Little's search for the flash. Say what changed Little's mind. Then draw or write one idea you want to keep.



No sun down here



The strong-stone surprise



Where the flash begins

Draw one detail from your scene. Add an arrow or a label to help someone understand it.

carbon · crystal · heat · pressure · scratch · light · facet

Deep underground, … What helped the diamond crystal grow?

A diamond is hard to scratch, but … What could still happen to it?

The flash begins when … What gives the light? How do the facets and watching eyes help?

Read it together. Point to the detail your words explain. If you are writing, check the first capital and the ending mark.